1 Linear Independence

Proposition 1.1. *Let V be a vector space and $\{v_1,\dots,v_m\}$ be a finite collection of vectors in V. Then, for each $v \in span(v_1,\dots,v_m)$,*

$$v = \alpha_1 v_1 + \cdots + \alpha_m v_m$$

for a unique *choice of scalars $\alpha_1,\dots,\alpha_m$ if and only if they only way to write 0 as a linear combination*

$$0 = \alpha_1 v_1 + \cdots + \alpha_m v_m$$

of $\{v_1,\dots,v_m\}$ is to take $\alpha_1 = \cdots = \alpha_m = 0$.

1. Prove Proposition 1.1.

Definition 1.2 (Linear independence)**.** A finite subset $\{v_1,\dots,v_m\}$ of vectors in a vector space V is *linearly independent* if the only way to write 0 as combination

$$\alpha_1 v_1 + \cdots + \alpha_m v_m = 0$$

is by taking $\alpha_1 = \cdots = \alpha_m = 0$. We also define $\emptyset$ to be linearly independent. If W is an infinite subset of V, then we define W to be linearly independent if every finite subset of W is linearly independent. If a subset of V is not linearly independent, then we say it is *linearly dependent*.

2. Prove that $\{(1,0,0),(0,1,0),(0,0,1)\}$ is linearly independent in F^3. More generally, the vectors e_i, defined as having 1 in the ith position and 0s elsewhere, are linearly independent in F^n.

3. Let m be a nonnegative integer. Prove that $\{1,t,\ldots,t^m\}$ is a linearly independent in $P(F)$, the vector space of all polynomials with coefficients in F.

Proposition 1.3.

(a) A singleton subset $\{v\}$ in a vector space V is linearly independent if and only if $v \neq 0$.

(b) A set of two vectors $\{v_1, v_2\}$ in a vector space V is linearly independent if and only if neither of the two vectors is a scalar multiple of the other.

4. Prove Proposition 1.3.

Proposition 1.4. *If $\{v_1, \ldots, v_m\}$ is linearly independent and we remove one or more vectors, then the set is still linearly independent.*

5. Prove Proposition 1.4.

6. Prove that $\{(2,3,1),(1,-1,2),(7,3,c)\}$ is linearly dependent in F^3 if and only if $c = 8$.

Proposition 1.5.

(a) If one vector in $\{v_1, \ldots, v_m\}$ is a linear combination of the other vectors, then $\{v_1, \ldots, v_m\}$ is linearly dependent.

(b) If one vector in $\{v_1, \ldots, v_m\}$ is the zero vector, then $\{v_1, \ldots, v_m\}$ is linearly dependent. (Note that this is a special case of part (a).)

7. Prove Proposition 1.5.

2 Exercises

1. Find a real number t such that $\{(3,1,4),(2,-3,5),(5,9,t)\}$ is not linearly independent in $\mathbb{R}^3$.

2. Consider the set $\{1+i, 1-i\} \subseteq \mathbb{C}$. Is this set linearly independent in $\mathbb{C}$ (viewed as a vector space over itself)? What if we think of $\mathbb{C}$ as a vector space over $\mathbb{R}$?

3. Prove or give a counterexample: If $\{v_1, v_2, v_3, v_4\}$ is linearly independent in V, then so is $\{v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4\}$.

4. Prove or give a counterexample: If $\{v_1, \ldots, v_m\}$ is linearly independent in V and $\lambda \in F \setminus \{0\}$, then $\{\lambda v_1, \ldots, \lambda v_m\}$ is linearly independent.

5. Prove or give a counterexample: If $\{v_1, \ldots, v_m\}$ and $\{w_1, \ldots, w_m\}$ are linearly independent in V, then $\{v_1 + w_1, \ldots, v_m + w_m\}$ is linearly independent.